Mapping the Spatial and Molecular Landscape of Lateral Hypothalamic Neurons Implicated in Eating

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INTRODUCTION

"Cell type" is defined in the contemporary transcriptomic sense: while cells were historically classified by morphology or electrophysiology, high-throughput molecular profiling has made gene-expression signatures a practical framework for identifying reproducible cellular identities, with the transcriptome serving as a proxy for underlying, evolutionarily encoded programs (Zeng, 2022). In this study, cell types are therefore operationalized as transcriptomically distinct groups discovered by unsupervised clustering of scRNA-seq data, with the central goal of identifying stable molecular identities within the lateral hypothalamic area (LHA) and determining how these identities are spatially organized. To do so, we combine data-driven clustering with functional annotation using established marker genes (e.g., Lepr) and then leverage MERFISH spatial transcriptomics to localize the annotated types within the LHA, enabling downstream questions about whether anatomically patterned transcriptomic identities relate to functional roles and behavioral phenotypes.

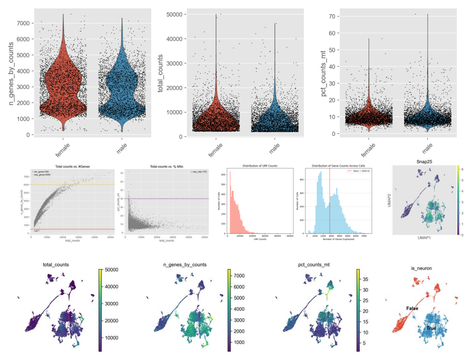
OBJECTIVES

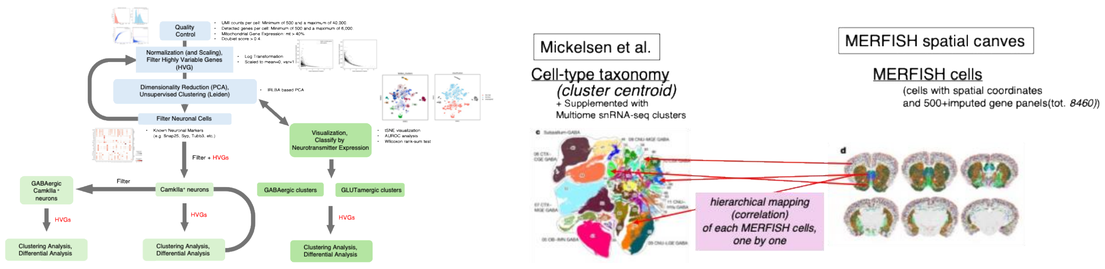
1. **Define stable LHA neuronal cell types:** Reconstruct an interpretable scRNA-seq–based taxonomy of LHA subtypes, highlighting molecularly distinct (and behavior-relevant) populations.
2. **Map cell types to MERFISH space :** Project the transcriptomic subtypes onto the Allen MERFISH atlas to visualize their spatial organization for functional, molecular annotation of the cell types.

METHODS

Dimension	MERFISH-imputed (ABC Atlas)	Mickelsen LHA scRNA-seq (2019)
Scope	• Whole-brain • single-cell • spatially resolved • integrated into the ABC taxonomy & portal.	Localized to LHA from microdissection (region-targeted dataset)
Measurement	Imaging-based molecule *spots per gene per cell* (panel genes); spatial coordinates for each cell.	UMI-based counts; LHA dissection & clustering pipeline specialized for LHA.
Gene coverage	500-gene MERFISH panel; + imputation of *8460 genes*	Hundreds–thousands of genes per cell; cluster-level markers validated (FISH & sc-qPCR).
Spatial coordinate system	Yes. (Registered to Allen CCFv3)	No
Cross-platform consistency	High cluster-level agreement with 10xv3 (median r≈0.91).	Unknown

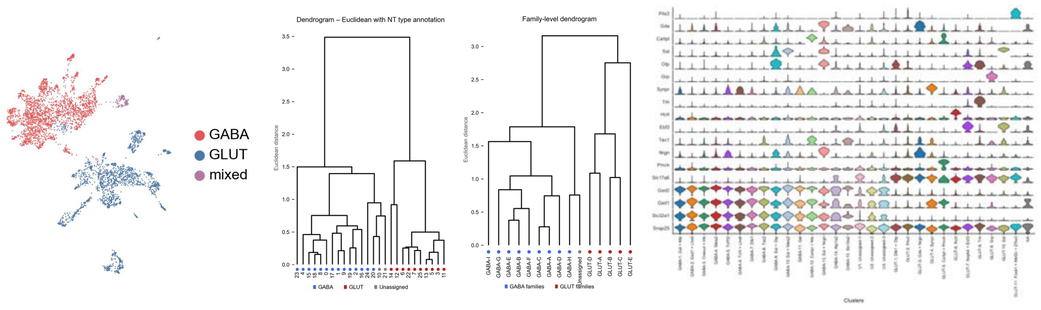
Hierarchical mapping (Yao, Zeng, Zhuang et al.)



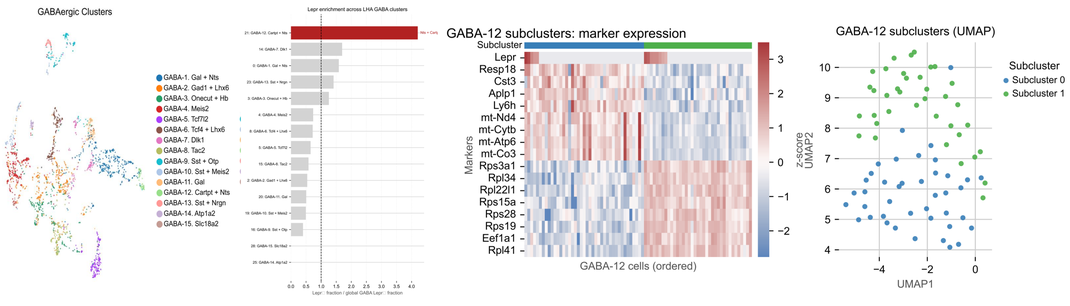


RESULTS

Clustering Analysis of the LHA neurons reveals a cell type hierarchy



[Example] A LepR-enriched GABAergic subtype in the LHA (GABA-12)



Mapping the Custom Taxonomy onto the ALLEN MERFISH Template

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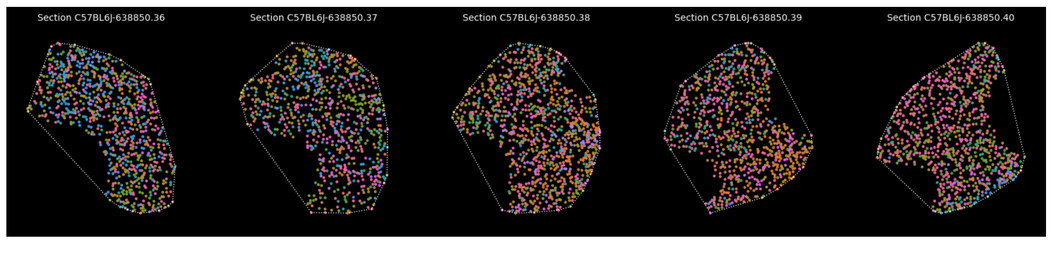
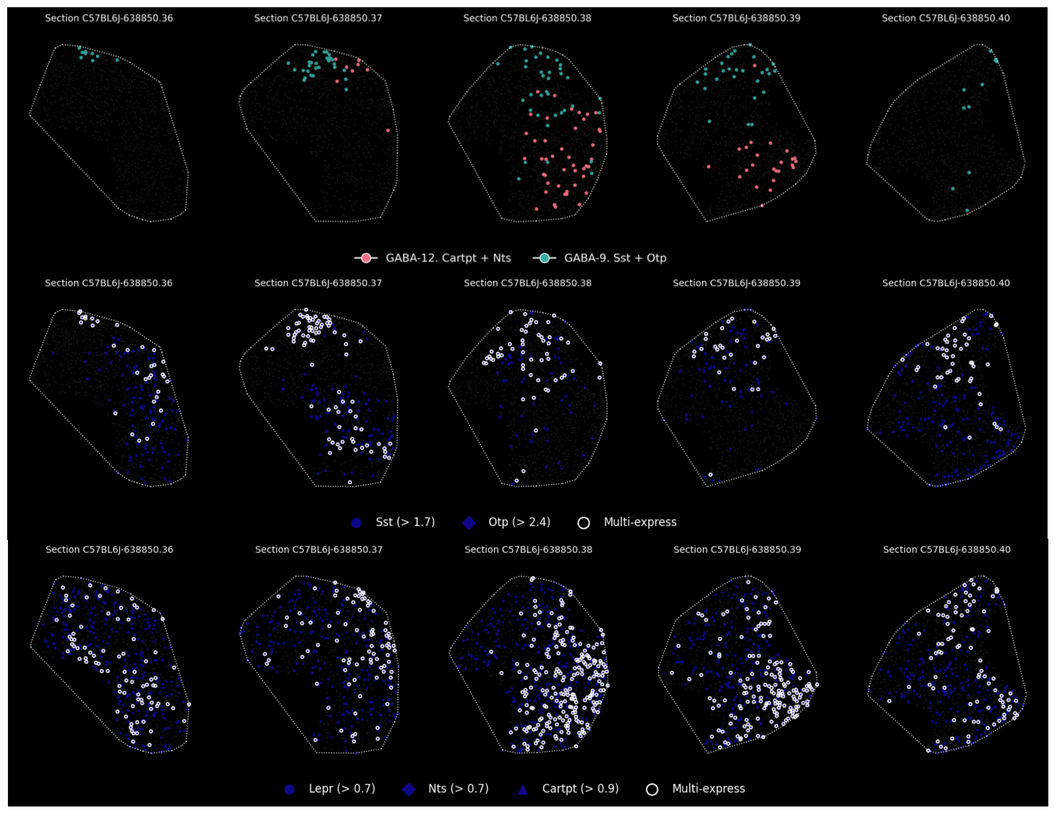
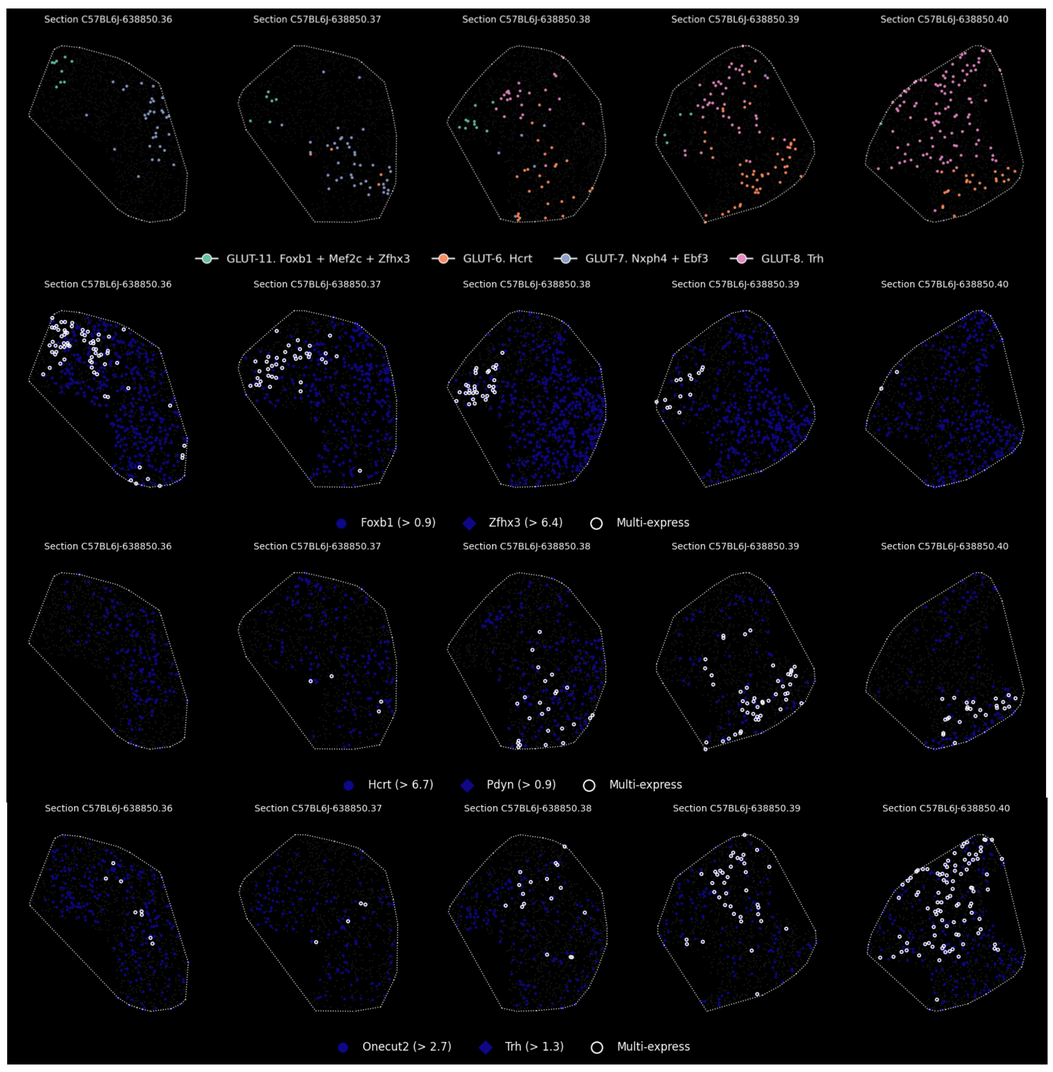


RESULTS

Spatial localization of GABA-12 versus GABA-9



Spatial localization of GABA-12 versus GABA-9



CONCLUSIONS

- **This study** builds a bridge from transcriptional diversity to spatial organization in LHA neurons by reanalyzing Mickelsen et al. scRNA-seq with a defined pipeline to derive an interpretable GABAergic vs glutamatergic subtype taxonomy, highlighting functionally interesting groups like **GABA-12 (Cartpt⁺/Nts⁺)** as strongly **Lepr-enriched**.
- Projects scRNA-defined subtypes onto the Allen MERFISH atlas via **centroid-based hierarchical matching**, revealing biased spatial distributions of certain subtypes; this yields approximate stereotaxic coordinates that can inform targeted experiments (e.g., viral injections, tracing, calcium imaging).
- **Key limitations when crossing modalities: Gene-panel mismatch** (MERFISH panel not optimized from scRNA DE markers) and only **modest centroid agreement** ($r \approx 0.67$) lead to imperfect mapping, distorted subtype abundances.
- **Practical takeaways:** Reliable subtype definitions require **marker panels optimized from DE analysis** and **high cross-modality concordance**, plus **filtering/down-weighting weak matches**; robust subtypes may need **iterative refinement** across datasets (re-DE, re-cluster with spatial priors, re-map) until stable.

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